CO21BTECH11002
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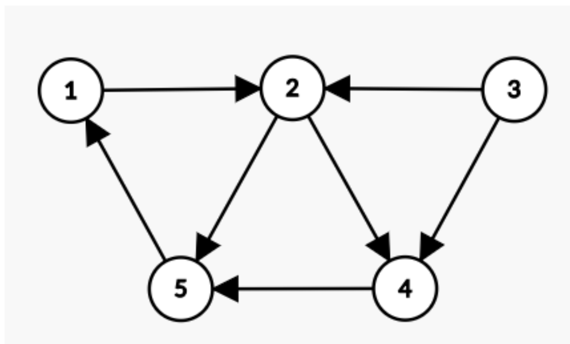
Theory Assignment 2

Q1 (3.4)

$s = r_1(x)r_3(x)w_3(y)w_2(x)r_4(y)c_2w_4(x)c_4r_5(x)c_3w_5(z)c_5w_1(z)c_1$

Conflict graph:



We can see that there is a cycle (1 -> 2 -> 5) in the graph. Hence, the history is not in CSR class.

For VSR and FSR, assuming there is a transaction t_0 that initialized the database and a transaction t_∞ that reads all the objects after all the operations are complete.

Then the read from for the given history:
$RF(s) = \{(t_0, x, t_1), (t_0, x, t_3), (t_3, y, t_4), (t_4, x, t_5), (t_4, x, t_\infty), (t_3, y, t_\infty), (t_1, z, t_\infty)\}$

From the $RF(s)$, we can conclude that the given history cannot be in VSR due to the following reason:

Suppose that there exists a sequential history s' such that s is VSR equivalent to s'.

Case 1:
In s', t_1 completes before t_5, then $w_1(z)$ will come before $w_5(z)$. Thus, $RF(s')$ will not contain (t_1, z, t_∞), since $r_\infty(z)$ will depend on the last write on z which can not be by t_1 in this case.

Case 2:
In s', t_5 completes before t_1. To maintain (t_4, x, t_5) in $RF(s')$, t_4 will have to complete before t_5 in s'. Now, in s', since t_4 completes before t_5 and in this case, we have assumed that t_5 completes before t_1, thus, t_4 completes before t_1. But then $w_4(x)$ will come before $r_1(x)$ and thus (t_0, x, t_1) will not appear in $RF(s')$ since $r_1(x)$ will depend on last write on x which cannot be t_0 if t_4 comes before t_1.

Hence, there is no case in which a valid sequential s' exists such that $RF(s) = RF(s')$.

Hence, the given history is not in VSR class.

Now, consider the live read from for the given history:
$LRF(s) = \{(t_0, x, t_1), (t_0, x, t_3), (t_3, y, t_4), (t_4, x, t_\infty), (t_3, y, t_\infty), (t_1, z, t_\infty)\}$

Let s' be a FSR equivalent sequential history.
To maintain (t_0, x, t_1) and (t_0, x, t_3) in $LRF(s')$, t_2 and t_4 must come after t_1 and t_3.
To maintain (t_3, y, t_4), t_4 must come after t_3.
To maintain (t_4, x, t_∞), t_4 must come after t_2.
No extra constraint is required to maintain (t_3, y, t_∞).
To maintain (t_1, z, t_∞), t_1 must come after t_5.

This is achievable and a valid sequence of transactions is: $t_5 t_1 t_3 t_2 t_4$
Hence, $s' = r_5(x)w_5(z)r_1(x)w_1(z)r_3(x)w_3(y)w_2(x)r_4(y)w_4(y)c_1c_2c_3c_4c_5$

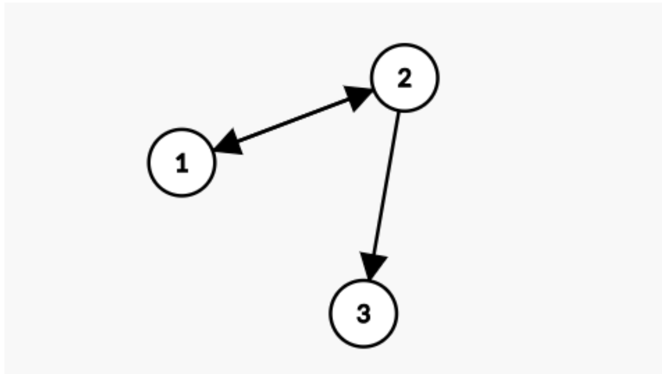
Also, $LRF(s) = LRF(s')$.

Hence, the given history s falls in FSR class.

Q2 (3.9)

$s = r_1(z)r_3(x)r_2(z)w_1(z)w_1(y)c_1w_2(y)w_2(u)c_2w_3(y)c_3$

Conflict graph:



There is a cycle in the conflict graph (1 -> 2 -> 1). Hence, s is not in CSR class.

Assuming there is a transaction t_0 that initialized the database and a transaction t_∞ that reads all the objects after all the operations are complete.

Read from for the given history:
$RF(s) = \{(t_0, z, t_1), (t_0, x, t_3), (t_0, z, t_2), (t_0, x, t_\infty), (t_3, y, t_\infty), (t_1, z, t_\infty), (t_2, u, t_\infty)\}$

Let s' be a sequential history such that $RF(s) = RF(s')$.

Then, to maintain (t_0, z, t_1) in $RF(s')$, no constraint is needed as only t_1 is writing on z.
To maintain (t_0, x, t_3), no constraint is needed as no one is writing on x.
To maintain (t_0, z, t_2), t_1 must write after t_2 has read z, so t_1 must come after t_2.
No constraint is required for (t_0, x, t_∞).
To maintain (t_3, y, t_∞), t_3 must write after t_1 and t_2 has written on y, so t_3 must come after t_1 and t_2.
No extra constraints are need for (t_1, z, t_∞) and (t_2, u, t_∞).

This is achievable and a valid sequence of transactions is: $t_2 t_1 t_3$.
Hence, $s' = r_2(z)w_2(y)w_2(u)r_1(z)w_1(z)w_1(y)r_3(x)w_3(y)$.

Also, $RF(s) = RF(s')$.

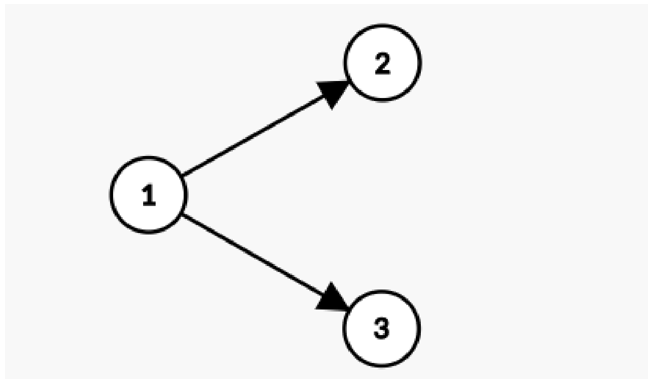
Hence, the given history s falls in VSR class but not in CSR class.

Hence, $s \in VSR - CSR$.

Q3 (3.10)

$s = r_1(x)w_1(z)r_2(z)w_1(y)c_1r_3(y)w_2(z)c_2w_3(x)w_3(y)c_3$

Conflict graph:



Since there is no cycle in the conflict graph, the given history does fall in CSR class.

A valid sequential history is:
$s' = t_1t_2t_3 = r_1(x)w_1(z)w_1(y)c_1r_2(z)w_2(z)c_2r_3(y)w_3(x)w_3(y)c_3$.

Also, for OCSR, in s, t_1 completes before t_2 and t_3 starts, and that happens in s' too. Also, t_2 and t_3 are interleaved in s so their completion order does not matter in s'.

Hence, s falls in OCSR class too.